

Supplementary Material: DIAGRARE-BENCH

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1 Supplementary Overview

The supplement gives the details needed to interpret and reproduce the measurements in the main paper. Section 2 describes the primary diagnostic cases and the two external case-report datasets. Section 3 expands the rank model and uncertainty procedure. Sections 4 and 5 define the randomized and sequential experiments. Section 6 records checkpoint coverage and inclusion rules. Section 7 tests alternative estimators, feature definitions, candidate sets, and negative controls. Sections 8 and 9 report model-level comparisons and interventions. Section 11 specifies the prompting protocol, and Section 12 summarizes the reproducibility materials.

2 Structured and External Evaluation Data

2.1 Primary diagnostic cases

The primary evaluation contains 984 cases covering 82 diagnoses and nine organ systems. Each diagnosis is represented by a disease profile containing required findings, supporting findings, findings whose absence is informative, and a disease-frequency attribute. The cases are generated from these profiles rather than from patient records. The evaluation is intentionally narrower than unrestricted diagnosis: the disease vocabulary is fixed, and the data specify which findings are present and absent. This structure is what allows the evidence and prior features to be defined independently.

For case x and candidate diagnosis d , the evidence score $e(d, x)$ summarizes agreement between the observed findings and the disease profile. Required findings receive larger weight than supporting findings, while explicit contradictions reduce the score. The primary scoring rule is fixed before comparing models. To test whether the conclusions depend on the particular weights, we recompute the evidence score over 16 parameter settings for required, supporting, and exclusionary findings. The diagnosis with the largest evidence score is unchanged in 99.9% of the 984 cases across this grid.

The disease-prior feature is derived from prevalence metadata. The main analysis uses a continuous log-prevalence value. Section 7.4 shows that replacing this representation with prevalence rank or with three rarity bins leaves the fitted evidence coefficient nearly unchanged for representative models.

2.2 External case-report evaluations

We evaluate all 3,562 CUPCase reports and all 22,901 RareArena cases. These datasets do not contain the structured disease-finding profiles used in the primary evaluation, so we construct an evidence feature directly from text. For each case, candidate diagnoses are scored by TF-IDF cosine similarity between the case presentation and the candidate diagnosis description. The prior feature is the log of one plus the PubMed article count for the diagnosis name. The text is truncated to 4,000 characters before both model inference and TF-IDF scoring so that the feature and the model receive the same case prefix.

The correct diagnosis has the largest lexical evidence score in 49.6% of CUPCase reports and 35.5% of RareArena cases. Both values exceed the 25% chance level of the four-candidate task. The external analysis therefore uses an evidence signal that is weaker than the structured feature but still informative. Since the numerical scales of the structured and lexical features are different, the paper compares model ordering across datasets rather than raw values of γ_{evidence} .

3 Rank Model and Estimation

3.1 Plackett-Luce likelihood

For model m , case x , and diagnosis d , define

$$\phi(d, x) = [p(d) \quad e(d, x)]^\top, \quad U_m(d, x) = \theta_m^\top \phi(d, x), \quad (1)$$

with $\theta_m = (\gamma_{\text{prior}}^{(m)}, \gamma_{\text{evidence}}^{(m)})^\top$. If the model returns the ordered list $d_1 \succ d_2 \succ \dots \succ d_K$ from candidate set $\mathcal{D}$, the likelihood is

$$P(d_1, \dots, d_K \mid x, \theta_m) = \prod_{k=1}^K \frac{\exp U_m(d_k, x)}{\sum_{d \in \mathcal{D} \setminus \{d_1, \dots, d_{k-1}\}} \exp U_m(d, x)}. \quad (2)$$

We estimate θ_m separately for each checkpoint with an L_2 penalty of 10^{-3} . For candidates a and b ,

$$\log \frac{P(a \succ b \mid x)}{P(b \succ a \mid x)} = \gamma_{\text{prior}} [p(a) - p(b)] + \gamma_{\text{evidence}} [e(a, x) - e(b, x)]. \quad (3)$$

The evidence coefficient is therefore the slope of fitted pairwise log odds with respect to an evidence difference. This interpretation is conditional on the feature scale and candidate set.

The rank model is not assumed to reproduce the network computation. It is a two-feature approximation to the output ordering of the model and decoder. Its usefulness depends on whether the fitted coefficient predicts behavior under independent evidence changes, which is why the randomized and sequential experiments are central to the paper.

3.2 Identifiability and protocol dependence

The estimator requires variation in both candidate features and enough valid ranked outputs to determine the two coefficients. A coefficient is not treated as an intrinsic property of a checkpoint. It can change with the prompt, candidate set, feature scale, and decoding rule. All within-benchmark model comparisons therefore use the same protocol. When the feature definition changes across datasets, we compare relative model ordering.

The method does not require a normalized probability distribution over diagnoses. It only requires an ordered list. This permits estimation from ordinary free-text differential diagnoses. Section 7 repeats the analysis with teacher-forced candidate scores for three models to verify that free-text parsing is not necessary for the main pattern.

3.3 Uncertainty

For the primary coefficient estimates, we bootstrap at the disease level so that all cases belonging to a resampled disease remain together. This choice treats disease identity as the principal semantic sampling unit. For representative models we also bootstrap at the organ-system level and compute leave-one-disease-out estimates. For Qwen2.5-7B, the disease-clustered 95% interval for γ_{evidence} is $[0.211, 0.251]$, the organ-clustered interval is $[0.211, 0.263]$, and the leave-one-disease-out range is $[0.228, 0.234]$. For Phi-3.5-mini, the corresponding intervals are $[0.328, 0.406]$ and $[0.324, 0.414]$, with a leave-one-disease-out range of $[0.360, 0.372]$.

4 Randomized Evidence and Stated-Prior Experiment

The rank coefficient is estimated from naturally varying cases. To test its interpretation more directly, we assign evidence strength and stated prevalence independently.

4.1 Disease pairs and assignment

We use 27 target and confounder disease pairs, with up to three pairs from each of the nine organ systems. Each pair is evaluated under three evidence levels and three stated-prior conditions, producing $27 \times 3 \times 3 = 243$ prompts per model. The stated-prior manipulation is intentionally large so that evidence and prior can be placed in clear conflict. In the target-rare condition, the prompt gives target and confounder rates of 1 and 10,000 per 100,000. The equal condition gives 1,000 and 1,000. The target-common condition reverses the first pair.

Evidence strength is generated from the target disease profile. Weak evidence contains one required target finding. Medium evidence contains approximately half of the required findings and one supporting finding. Strong evidence contains all required findings and up to two supporting findings; where applicable, it also states the absence of findings that define the confounder. These prompts are not part of the data used to fit baseline γ_{evidence} .

4.2 Finite-difference estimands

Let $Y_{m,j}(e, p)$ denote target choice for model m , pair j , evidence level e , and stated-prior condition p . The model-level evidence effect is

$$\Delta_E^{(m)} = \frac{1}{3J} \sum_{j=1}^J \sum_p [Y_{m,j}(e_{\text{strong}}, p) - Y_{m,j}(e_{\text{weak}}, p)]. \quad (4)$$

The stated-prior effect is

$$\Delta_P^{(m)} = \frac{1}{3J} \sum_{j=1}^J \sum_e [Y_{m,j}(e, p_{\text{target common}}) - Y_{m,j}(e, p_{\text{target rare}})]. \quad (5)$$

We also compute the evidence-by-prior interaction and the strong-evidence response under the unfavorable target-rare condition. Confidence intervals resample disease pairs.

Across the 21 models with baseline estimates, γ_{evidence} predicts Δ_E with Pearson $r = 0.689$ ($p = 5.5 \times 10^{-4}$) and Spearman $\rho = 0.668$. The matched relationship between γ_{prior} and Δ_P is weak and not significant ($r = -0.143$, $p = 0.535$). The evidence coefficient is therefore the quantity for which the intervention supports the intended interpretation. We use the prior coefficient as a conditioning coordinate and do not make a symmetric validity claim for it.

Random assignment supports a causal statement about the effect of changing the prompt evidence on the model output within this experiment. It does not establish an internal causal representation or human-like causal reasoning.

5 Sequential Diagnostic Revision

The factorial experiment compares completed prompts. We next measure whether a model revises a diagnosis when evidence arrives in two steps. The experiment contains 82 target and confounder pairs. Each pair is presented in two orders. In the confounder-first order, the first block supports the confounder and the second adds target-specific findings. In the target-first order, the same two blocks are reversed. The union of evidence at step two is identical across orders.

Let $Y_{m,j,t}^{C \rightarrow T}$ and $Y_{m,j,t}^{T \rightarrow C}$ be binary target-choice indicators. The recovery rate is

$$R_m = \frac{\sum_j \mathbf{1}[Y_{m,j,1}^{C \rightarrow T} = 0] Y_{m,j,2}^{C \rightarrow T}}{\sum_j \mathbf{1}[Y_{m,j,1}^{C \rightarrow T} = 0]}. \quad (6)$$

The final-step order effect is

$$O_m = \frac{1}{J} \sum_j (Y_{m,j,2}^{C \rightarrow T} - Y_{m,j,2}^{T \rightarrow C}). \quad (7)$$

Seventeen models have at least five cases in the denominator of R_m . Baseline γ_{evidence} predicts recovery with Pearson $r = 0.860$ ($p = 9.8 \times 10^{-6}$) and Spearman $\rho = 0.811$. All 17 models have a positive mean order effect. Under the sign convention above, this means target evidence has greater influence when it appears second. We therefore describe the effect as recency or path dependence rather than persistent anchoring.

6 Model Panel and Inclusion Rules

The model panel contains 25 open-weight checkpoints. It deliberately includes both general-purpose models and models specialized for medicine, since a diagnostic benchmark should not assume that medical specialization is the only route to strong performance. Both groups receive the same diagnostic prompts, candidate vocabulary, scoring rules, and rank estimator. Twenty-one checkpoints provide enough valid ranked outputs to identify the two coefficients in the primary

analysis. Later experiments use different response formats, so their model counts are reported separately rather than forcing all comparisons onto one reduced subset.

Table 1 lists the model panel. The table is intentionally descriptive: it records checkpoint identity, model family, and parameter count, while experiment-specific coverage is reported with the analysis that uses it. This keeps the model panel separate from the statistical inclusion rules and avoids attaching qualitative labels to incomplete outputs.

Table 1: Open-weight checkpoints in the model panel. General-purpose and medically specialized checkpoints are evaluated with the same diagnostic protocol.

Checkpoint	Family	Params. (B)
General-purpose checkpoints		
Qwen2.5-0.5B	Qwen	0.5
Qwen2.5-1.5B	Qwen	1.5
Qwen2.5-3B	Qwen	3.0
Qwen2.5-7B	Qwen	7.0
Mistral-7B-Instruct	Mistral	7.3
Phi-3.5-mini	Phi	3.8
Yi-1.5-6B	Yi	6.0
Yi-1.5-9B	Yi	9.0
OLMo-2-7B	OLMo	7.0
DeepSeek-LLM-7B	DeepSeek	7.0
Falcon-7B	Falcon	7.0
StableLM-Zephyr-3B	StableLM	3.0
Granite-3.1-2B	Granite	2.0
Granite-3.1-8B	Granite	8.0
Zephyr-7B	Mistral	7.2
OpenChat-3.5	Mistral	7.0
Vicuna-7B	Llama2	7.0
TinyLlama-1.1B	Llama	1.1
GLM-4-9B	GLM	9.0
Nous-Hermes-2-7B	Mistral	7.0
Medically specialized checkpoints		
MedAlpaca-7B	Llama2	7.0
OpenBioLLM-8B	Llama3	8.0
Asclepius-7B	Mistral	7.0
Med42-8B	Llama3	8.0
BioMistral-7B	Mistral	7.0

A checkpoint enters the primary coefficient analysis when the two-parameter likelihood is identifiable and there are enough valid ranked responses to estimate the coefficient. We do not impute missing experiments. Every cross-model statistic therefore reports its own model count. This matters most for checkpoints whose free-text output does not consistently follow the requested ranking format.

Because low output coverage can increase variance, we repeat the principal association after imposing stricter coverage thresholds. Requiring at least 50% valid rankings leaves 17 models and increases the γ_{evidence} -accuracy correlation from $r = 0.843$ to $r = 0.974$. Thresholds at 25% and 10% maximum failure give the same qualitative conclusion. A descriptive model-level regression that includes γ_{evidence} , log parameter count, output-failure rate, and model-family fixed effects retains a significant coefficient on γ_{evidence} ($p = 0.0028$). These checks are not used to redefine the panel; they test whether the cross-model relationship is driven by a small set of poorly formatted outputs.

Table 2: Representative estimates from the primary ranking protocol. Top-1 accuracy and the two fitted coefficients are shown for checkpoints spanning the observed range of evidence responsiveness. The complete table is included in machine-readable form.

Checkpoint	Type	Top-1 (%)	γ_{prior}	γ_{evidence}
Qwen2.5-0.5B	General	0.1	-0.107	-0.072
Qwen2.5-1.5B	General	6.2	0.007	0.076
Qwen2.5-3B	General	34.7	0.043	0.212
Qwen2.5-7B	General	41.1	0.059	0.230
OLMo-2-7B	General	53.0	0.053	0.280
Mistral-7B-Instruct	General	62.4	0.112	0.325
Phi-3.5-mini	General	69.8	0.150	0.362
Med42-8B	Medical	70.0	0.098	0.342
MedAlpaca-7B	Medical	2.4	0.381	0.018

7 Robustness of the Rank Model

7.1 Alternative choice likelihood

The main estimator uses Plackett-Luce because the model returns an ordered list. We also fit a Bradley-Terry pairwise likelihood to eight representative checkpoints. The absolute scale changes under the different likelihood, but the ordering is similar. For Qwen2.5-1.5B, Qwen2.5-3B, Qwen2.5-7B, OLMo-2-7B, Mistral-7B-Instruct, and Phi-3.5-mini, the Bradley-Terry evidence coefficients are 0.086, 0.196, 0.238, 0.246, 0.280, and 0.294, respectively. The sequence agrees with the increase seen under Plackett-Luce.

7.2 Teacher-forced candidate scoring

Free-text parsing could confound the analysis if checkpoints with cleaner formatting are also more accurate. For three checkpoints with accessible logits, we score candidate diagnoses directly under teacher forcing and regress the resulting score differences on prior and evidence. The direct evidence coefficient is positive for Qwen2.5-1.5B (0.019, $p = 1.5 \times 10^{-8}$), Mistral-7B-Instruct (0.127, $p < 10^{-200}$), and Phi-3.5-mini (0.065, $p < 10^{-180}$). These coefficients are not numerically comparable to the ranking coefficients because the dependent variable is now a logit-derived score, but the result removes free-text parsing from the measurement.

7.3 Feature-shuffling test

We independently permute one feature within each case and refit the rank model. Shuffling the evidence score reduces γ_{evidence} from 0.230 to 0.003 for Qwen2.5-7B, from 0.362 to 0.008 for Phi-3.5-mini, from 0.325 to -0.012 for Mistral-7B-Instruct, and from 0.212 to 0.003 for Qwen2.5-3B. Shuffling the prior leaves γ_{evidence} nearly unchanged. The selective collapse is consistent with the two features contributing distinct information to the fitted ranking.

7.4 Prevalence representation

The continuous prevalence representation is not necessary for the evidence result. Across eight representative models, replacing continuous prevalence with within-set rank or three rarity bins changes γ_{evidence} only slightly. For Qwen2.5-7B the values are 0.230, 0.229, and 0.229; for Phi-3.5-mini they are 0.362, 0.361, and 0.361. The prior coefficient changes more because the feature scale has been changed directly, which is why we do not compare the magnitude of γ_{prior} across encodings.

7.5 Candidate-set dependence

Choice coefficients depend on the alternatives available to the model. We recompute the rank fit using the full 82-disease set, candidates from the same organ system, and the same-organ set augmented with globally common diseases.

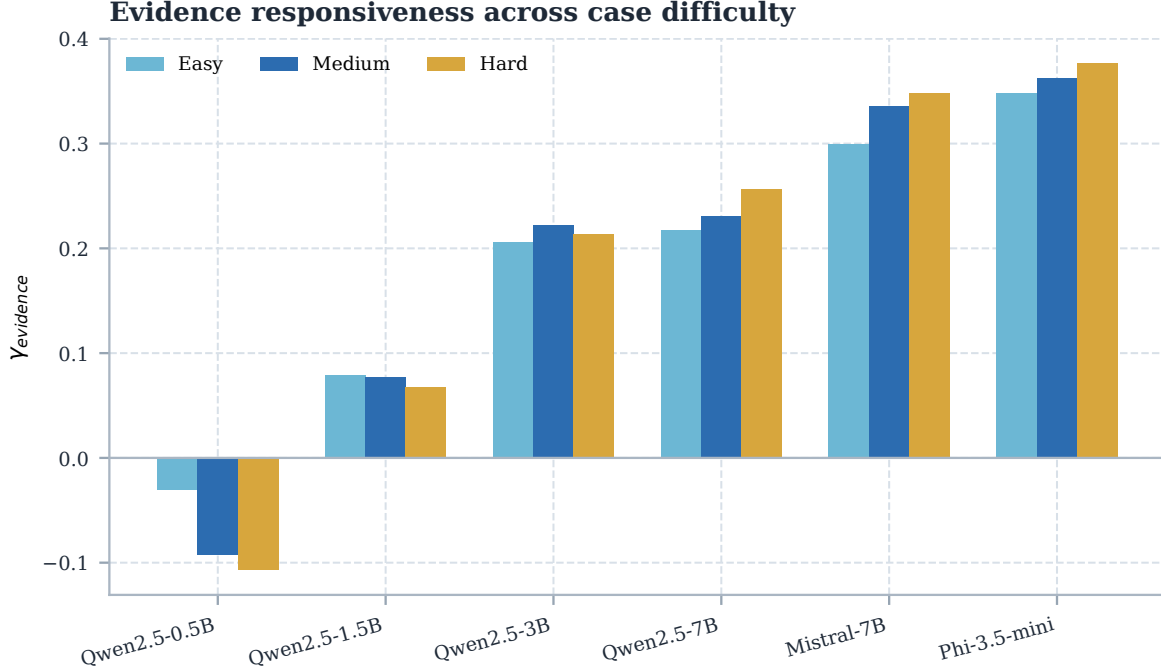


Figure 1: Evidence responsiveness stratified by case difficulty.

Absolute coefficients shrink under the restricted sets, but the ordering of representative models is broadly preserved. Qwen2.5-7B has evidence coefficients 0.230, 0.155, and 0.160 across the three candidate sets; Phi-3.5-mini has 0.362, 0.239, and 0.245. This dependence is expected and is the reason we interpret γ_{evidence} comparatively under a fixed evaluation protocol.

7.6 Irrelevant-text negative control

A checkpoint that changes its answer in response to any prompt edit could appear evidence responsive without using diagnostic content selectively. We append one clinically irrelevant sentence to every primary case and measure the top-1 flip rate. Across eight models, baseline γ_{evidence} is negatively correlated with irrelevant-text sensitivity ($r = -0.753$, $p = 0.031$). Models with larger evidence coefficients are therefore not simply more unstable under textual changes.

7.7 Matched rare and common targets

Rare cases could mechanically appear less evidence responsive if they contain fewer or weaker findings. We form 432 rare and common matched pairs controlling organ system, difficulty, number of findings, and target evidence. Mean finding counts are 9.46 for rare cases and 9.35 for common cases; mean target-evidence scores are 16.10 and 15.70. Six of eight tested models have lower γ_{evidence} on the rare subgroup, Phi-3.5-mini is approximately equal, and Qwen2.5-0.5B is unstable because its task performance is near floor. The result is therefore not explained simply by rare cases containing less information.

7.8 Sensitivity across evidence parameters and organ systems

We fit organ-specific coefficients and a hierarchical shrinkage model to check whether the global result is dominated by one clinical domain. The ordering remains visible across organ systems, although smaller strata have wider uncertainty. We also sweep the evidence-score parameters and recompute the model-level association over the full grid. The main relationship remains stable across this range.

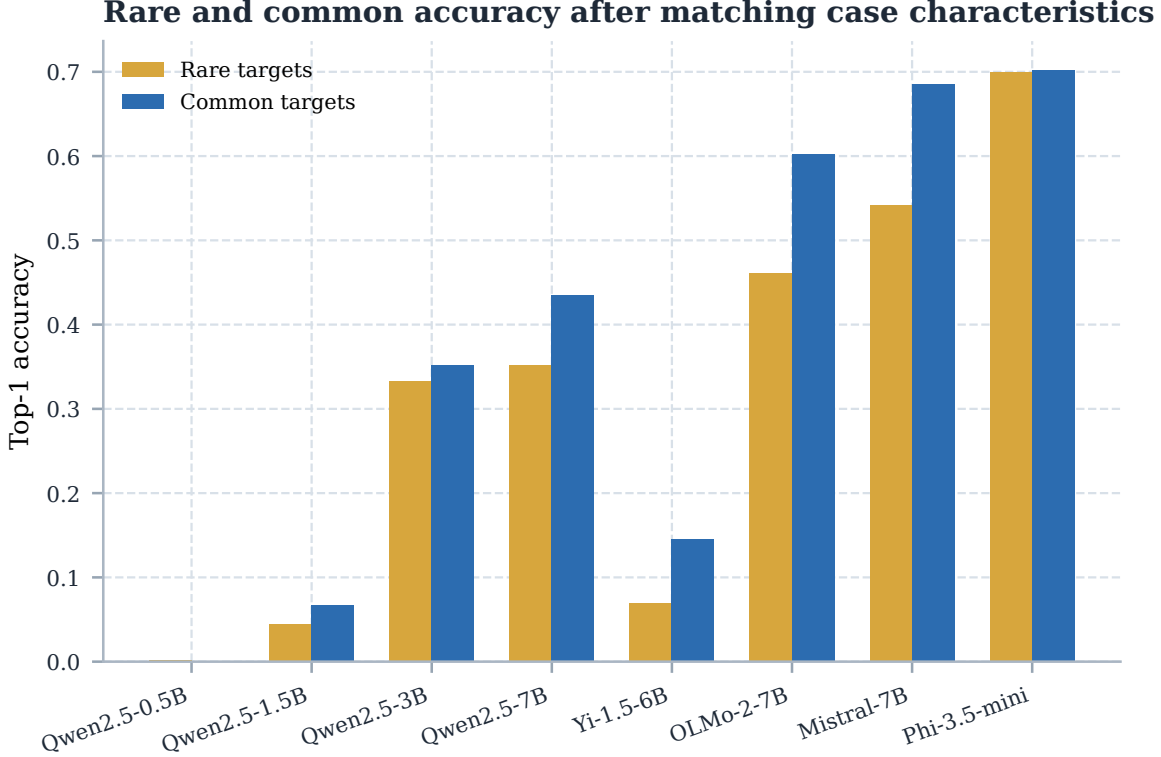


Figure 2: Rare and common accuracy after matching organ system, difficulty, finding count, and target evidence.

8 Additional Model-Level Analyses

8.1 Parameter count and medical specialization

Parameter count alone does not explain the main cross-model relationship. A regression using only log parameter count over 25 models has $R^2 = 0.109$ and $p = 0.107$. In the 21-model regression that contains γ_{evidence} , log parameter count, parsing failure rate, and model-family fixed effects, γ_{evidence} remains significant ($p = 0.0028$), while log parameter count does not ($p = 0.886$). These comparisons are observational and are not interpreted as causal effects of scale.

We also compare medically specialized and general checkpoints in a restricted panel. After accounting for γ_{evidence} and parameter count, the specialization indicator is not significant ($p = 0.242$). The available medically specialized checkpoints differ in base architecture, data, and training procedure, so this comparison is confounded and underpowered. We report it only as a check that the main result is not produced by a simple specialized-versus-general partition.

9 Model and Inference Interventions

9.1 QLoRA adaptation

We adapt Qwen2.5-7B with QLoRA on 72 examples spanning six diseases. The training examples use the same ranked-diagnosis format as the primary evaluation but do not include the randomized or sequential prompts. On the original 696-case split, top-1 accuracy increases from 40.9% to 63.1%, while γ_{evidence} increases from 0.215 to 0.237. We also evaluate 216 cases from organ systems absent from the adaptation set. Accuracy increases from 43.1% to 63.0%, and γ_{evidence} increases from 0.306 to 0.386. The effect therefore transfers beyond the organ systems represented in the adaptation examples.

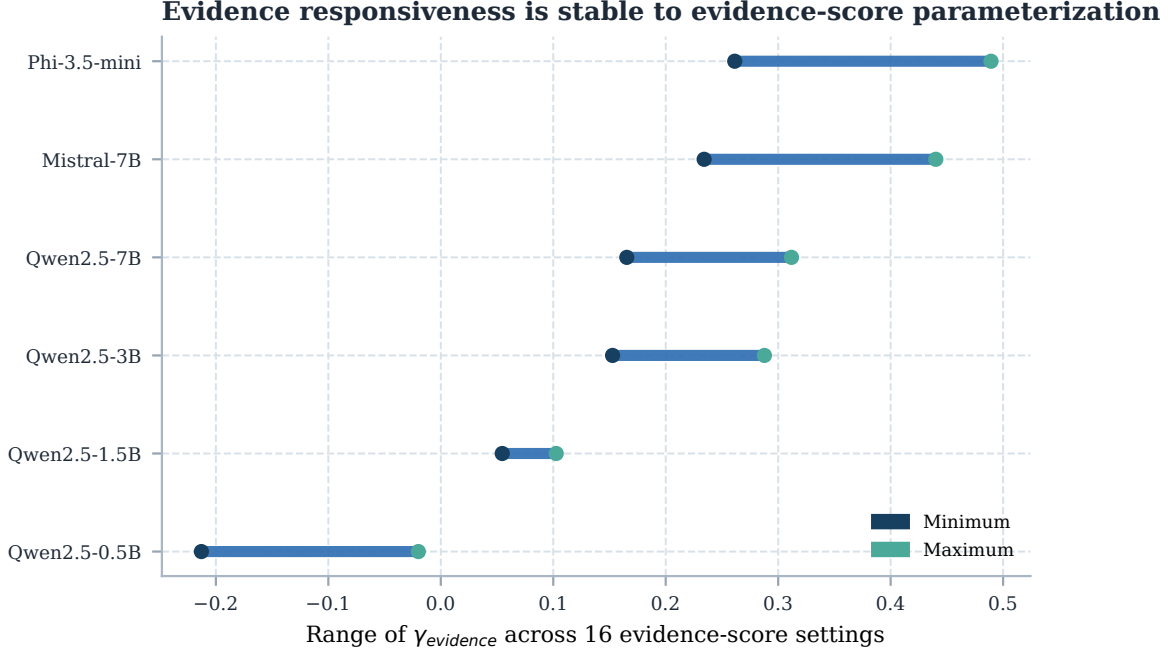


Figure 3: Sensitivity of the cross-model result to alternative evidence-score parameterizations.

9.2 Self-consistency

For three Qwen2.5 checkpoints, we sample five independent ranked outputs and aggregate them. The 0.5B model remains near floor. For Qwen2.5-3B, top-1 accuracy increases from 34.7% to 41.3%, while γ_{evidence} changes from 0.212 to 0.222. For Qwen2.5-7B, accuracy increases from 41.1% to 47.5%, while γ_{evidence} changes from 0.230 to 0.244. Thus, aggregation improves final correctness more than it changes the estimated dependence of the ranking on patient evidence.

9.3 Prompt and chat-template sensitivity

We evaluate alternative instruction wording and chat-template application for representative checkpoints. A poorly matched template can reduce absolute accuracy, but the evidence-response pattern is stable under the intended chat template for each checkpoint. These runs are diagnostics of protocol sensitivity; prompts are not tuned separately to maximize each model’s score.

9.4 Qwen3 reasoning budget

Qwen3-8B supports thinking and no-thinking modes within the same checkpoint, which permits a within-model comparison. With thinking disabled, the model completes nearly all primary ranking prompts and obtains 74.4% top-1 accuracy, $\gamma_{\text{evidence}} = 0.334$, $\Delta_E = 0.309$, and sequential recovery of 0.975. Under a 512-token thinking budget, all 984 primary ranking prompts exhaust the budget and only four produce usable rankings. Under a 2,048-token budget, 89.3% of 300 evaluated primary prompts exhaust the budget and 47 yield usable rankings. The resulting baseline coefficients are therefore not directly comparable to the no-thinking estimate because the usable case sets differ substantially.

The binary intervention tasks require less output. Their evidence effects are 0.309, 0.000, and 0.136 for no-thinking, 512-token, and 2,048-token conditions; recovery rates are 0.975, 0.983, and 0.962. Within this protocol, the tested reasoning budgets do not increase evidence responsiveness and introduce a large completion cost on the 82-way ranking task. The result is limited to this checkpoint, prompt format, and budget range.

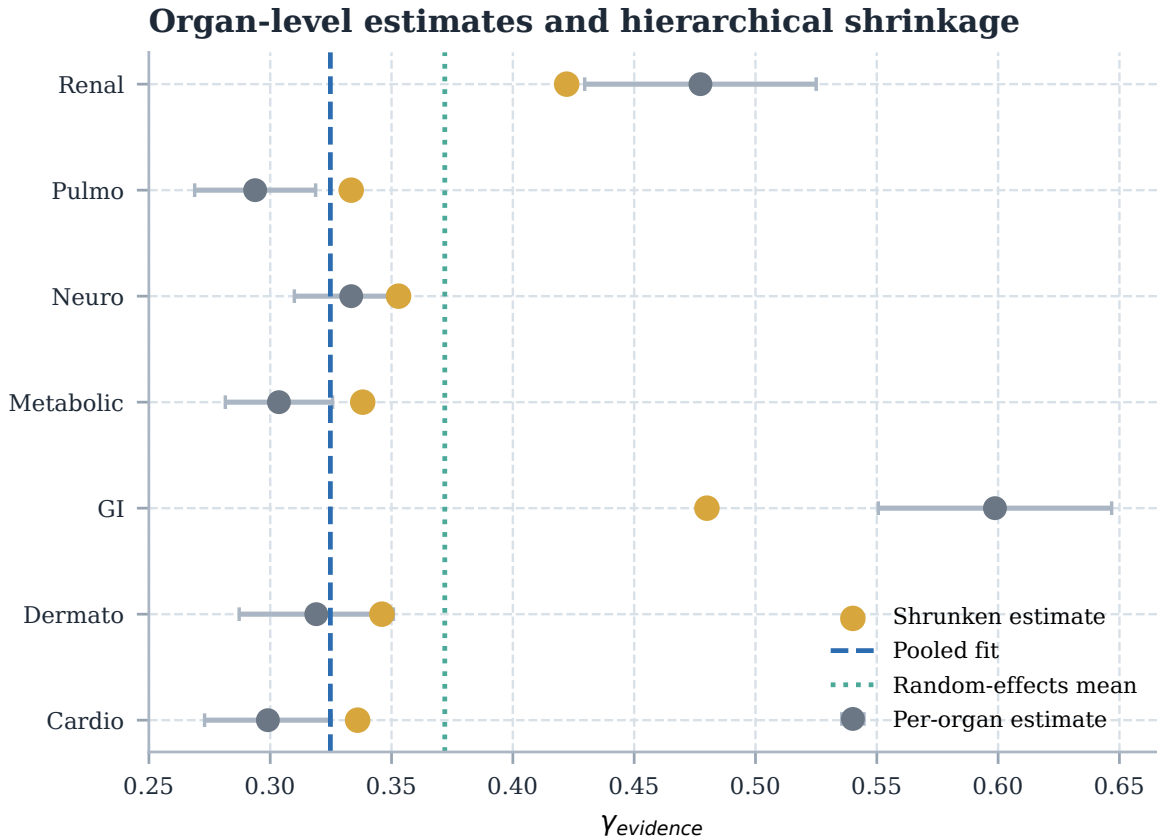


Figure 4: Organ-level estimates and hierarchical shrinkage.

10 Prior and Evidence Coefficient Landscape

Figure 9 plots the two fitted coefficients for the open-weight panel. The plot is descriptive and is included to show how the two fitted coordinates vary across checkpoint families. The evidence coefficient has direct interventional validation; the prior coefficient is retained as the second coordinate of the rank model and does not receive the same interpretation.

11 Prompt Protocols

11.1 Primary ranking prompt

The primary prompt presents positive findings, explicitly absent findings, and the fixed disease vocabulary. It asks for the three most likely diagnoses in ranked order and requests disease names only. Model-specific chat templates are used when provided by the checkpoint. No chain-of-thought explanation is requested or scored.

A representative prompt is:

Given the clinical findings below, return the three most likely diagnoses from the supplied disease list. Use both the positive findings and the explicitly absent findings. Return a numbered list of three disease names, ordered from most to least likely.

The exact case findings and candidate list are inserted by the included code.

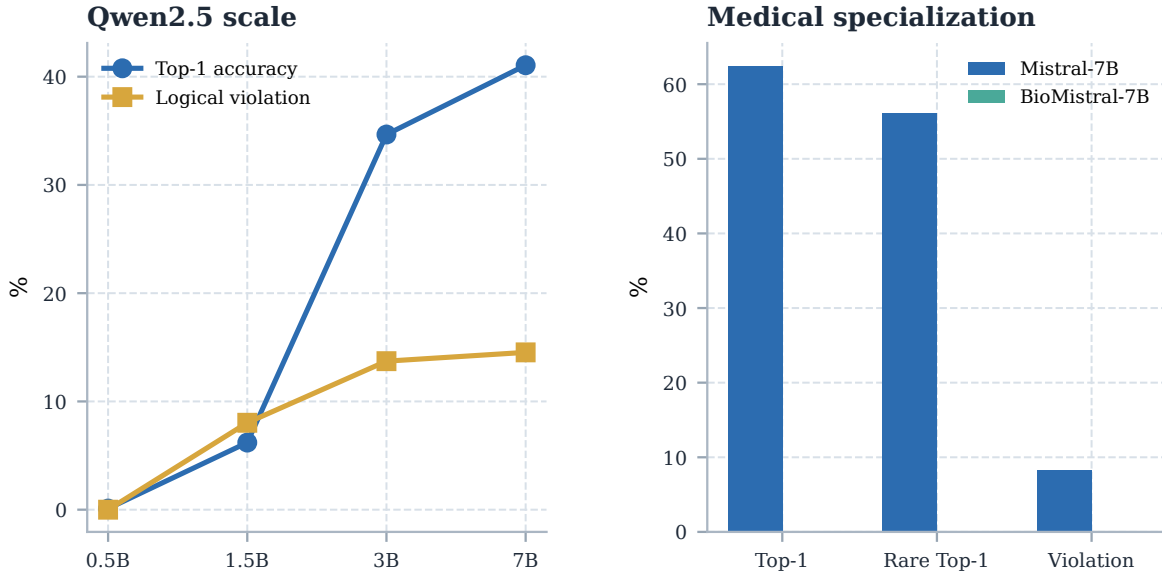


Figure 5: Descriptive analyses of parameter count and medical specialization.

11.2 Randomized intervention prompt

The factorial prompt presents two diagnoses, an assigned statement about their prevalence, and findings corresponding to the assigned evidence level. The model must choose one diagnosis. Evidence and stated prevalence are assigned independently, so the prompt can pair strong target evidence with an unfavorable target prior.

11.3 Sequential prompt

The sequential task preserves conversational state. At step one, the model receives the first evidence block and returns a diagnosis. At step two, it receives the second block and is asked to update the diagnosis using all information seen so far. The reverse-order condition swaps the two blocks but gives the same union of evidence at the second step.

12 Reproducibility

The benchmark data and analysis code are available in the supplementary appendix and will be released publicly upon acceptance.

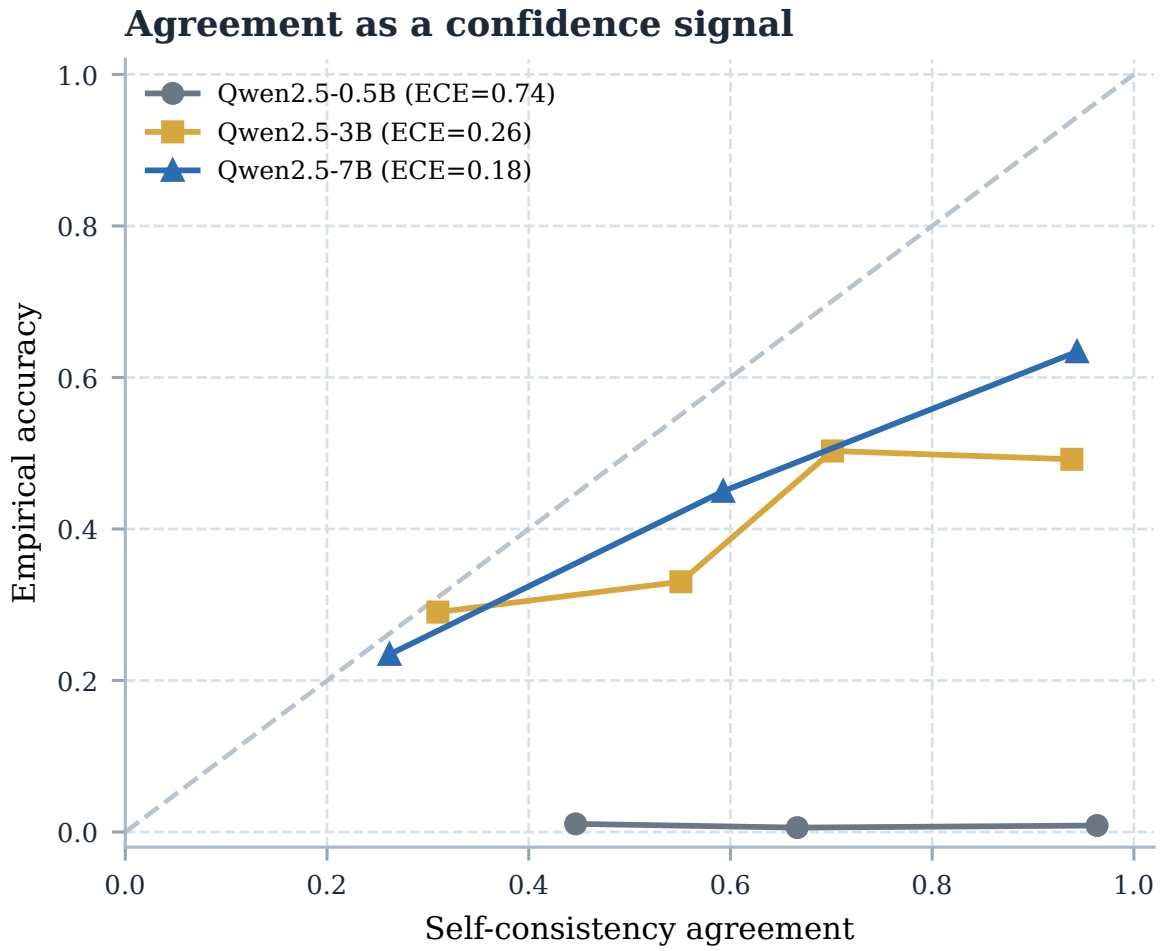


Figure 6: Self-consistency agreement as a confidence signal for the Qwen2.5 checkpoints.

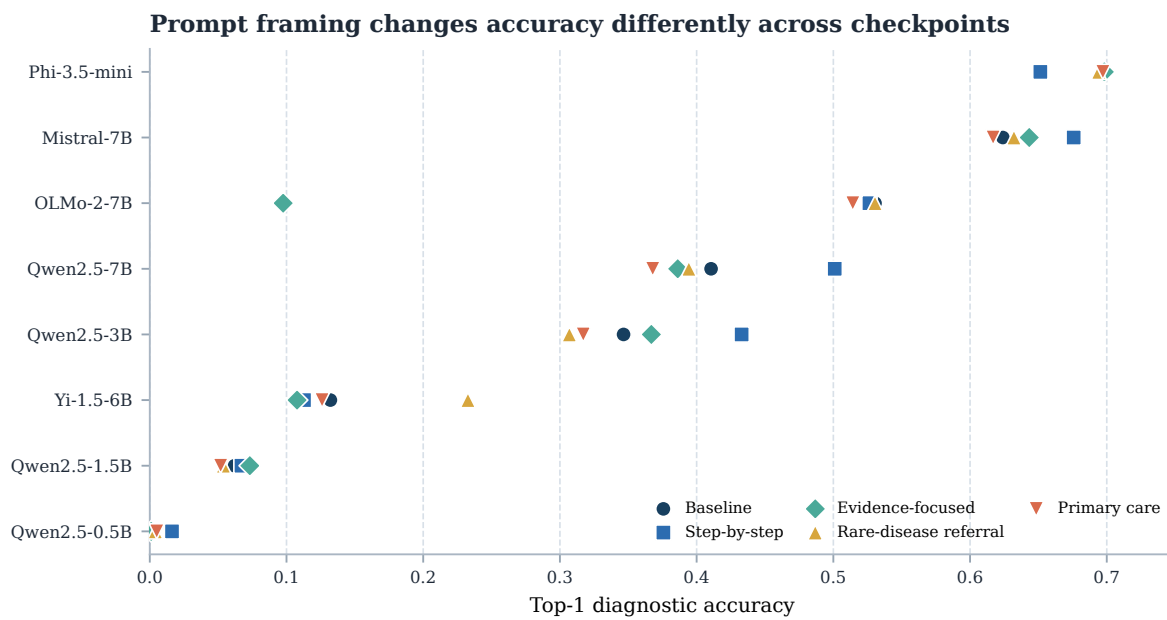


Figure 7: Prompt and decoding sensitivity across the evaluated checkpoints.

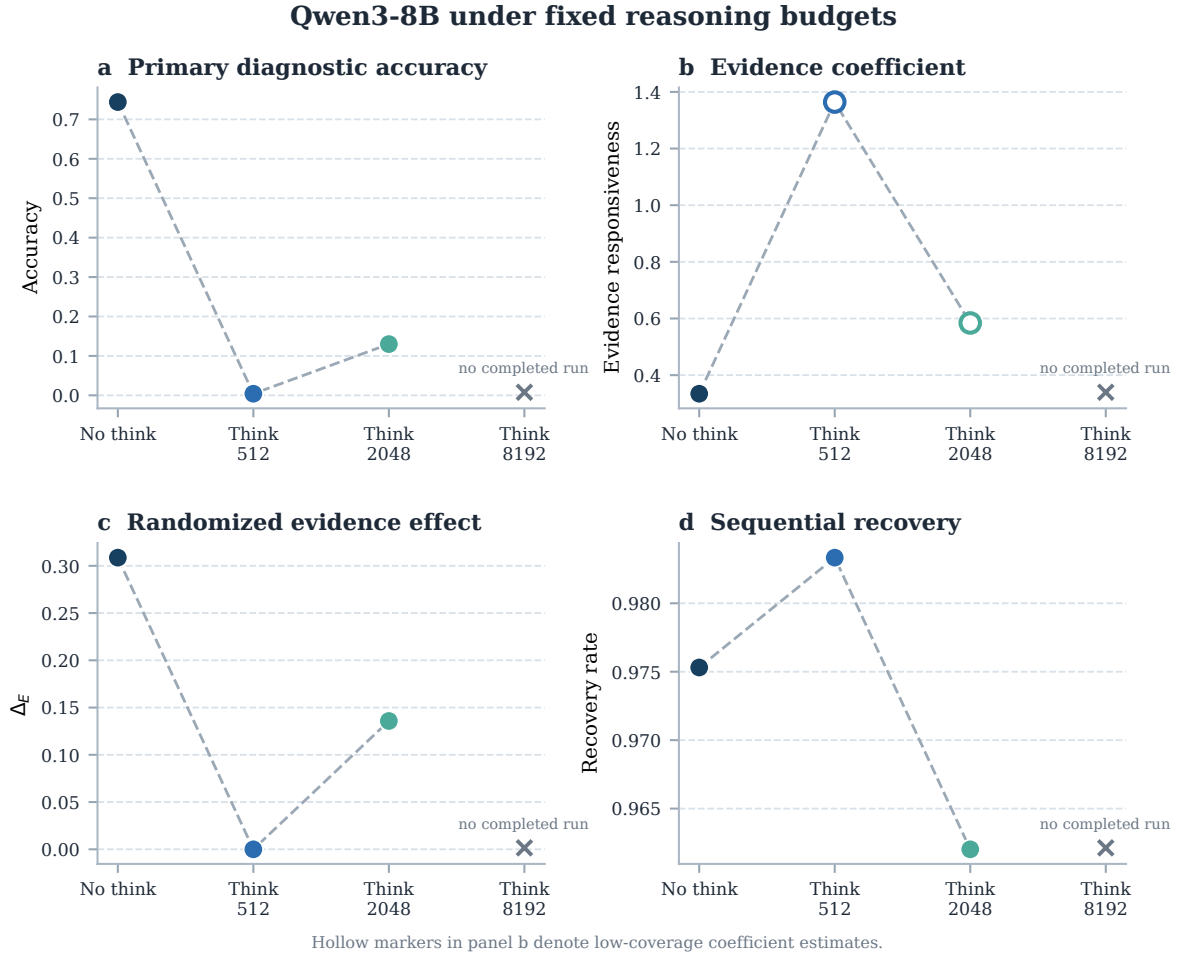


Figure 8: Qwen3-8B under fixed reasoning budgets. Baseline ranking coefficients become difficult to compare when the reasoning trace consumes the available budget before the requested ranked answer is produced.

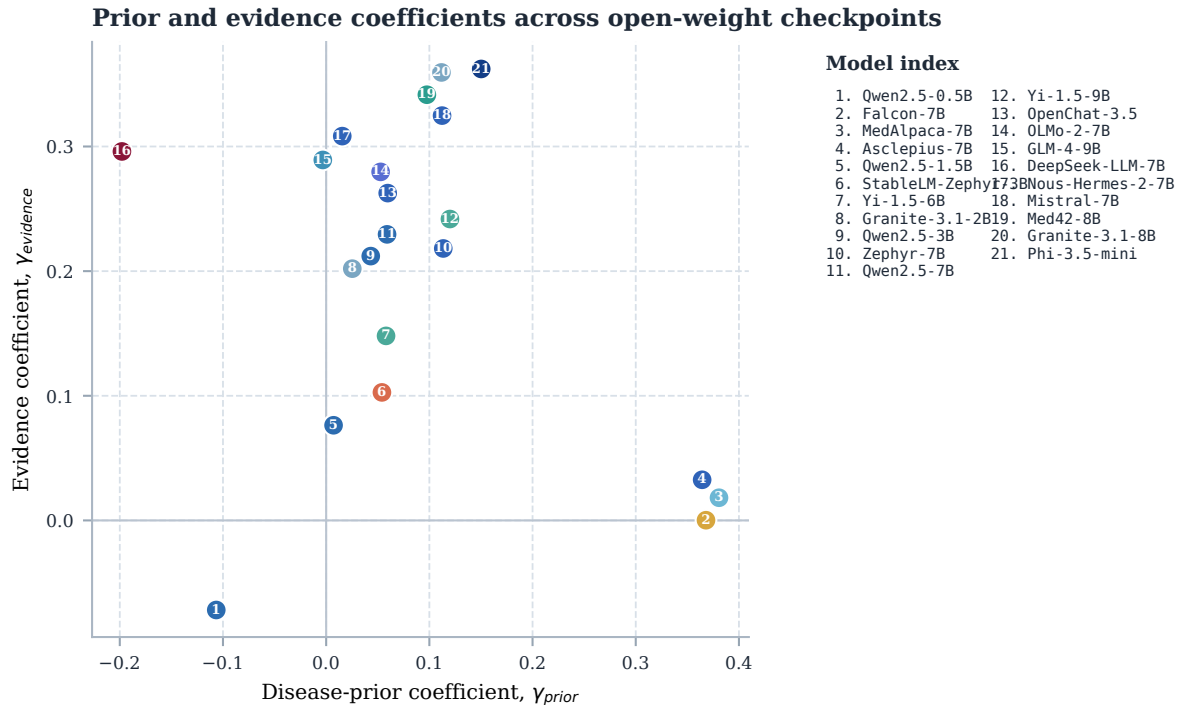


Figure 9: Prior and evidence coefficients across the open-weight model panel.